

A Goal-Conditioned Information-Theoretic Framework for Modeling Human Attention

Abstract

Background/Objective: Human attention is often described as focus or concentration, but modern cognitive environments require a more precise measurement language. This paper asks whether aspects of attention can be modeled as goal-conditioned information transfer rather than as simple time-on-task. **Methods:** A conceptual framework was developed from cognitive psychology, information theory, and cognitive control research. A proof-of-concept analysis used a public Stroop dataset with 131 participant rows. Congruent and incongruent block completion times were compared with a paired-samples t-test. Response-time values were also divided into pooled tertile bins, and mutual information was used to estimate how much task-condition information was preserved in response-time bins. **Results:** Congruent blocks were completed faster than incongruent blocks, producing a mean interference effect of 7.90 s, paired $t(130) = 18.04$, $p < .001$, Cohen's $d_z = 1.58$. The contingency analysis showed raw $I(C;B_RT) = 0.404$ bits, Miller-Madow corrected $I(C;B_RT) = 0.399$ bits, and a bootstrap 95% interval of $[0.324, 0.505]$ bits. **Conclusions:** The results demonstrate that an information-theoretic calculation can complement traditional response-time analysis, but the demonstrated quantity is condition information in response-time bins, not full attentional bandwidth. A stricter test would require trial-level behavioral, eye-tracking, or neural data with preregistered relevance partitions.

Keywords: attention; information theory; mutual information; Stroop task; cognitive control; goal-conditioned relevance; artificial intelligence

Introduction

Human attention has been treated as a limited cognitive resource for more than half a century. Early filter and attenuation theories described selective processing as a bottleneck, capacity theories framed attention as controlled effort, and network theories separated alerting, orienting, and executive control. These traditions agree that cognition cannot process every available stimulus with equal depth, but they differ in what the limiting resource is and how it should be measured^{1,2,3,4}.

This paper begins from a narrower measurement question: can aspects of attention be represented with the same mathematical seriousness used to describe information channels? In communication theory, entropy and mutual information quantify uncertainty reduction between variables. These tools do not explain meaning by themselves, but they can describe whether one variable statistically preserves information about another^{7,8}.

The central claim is not that attention is literally internet bandwidth. That would be a reductionist metaphor. The claim is that some attentional operations can be modeled as goal-conditioned information control. A person or system has a goal G , an action space, and a loss function $L(a,s)$. Under that goal, some stimulus features become relevant because changing them would change the correct action, expected loss, or task success. Other features may still be statistically salient while remaining irrelevant to the goal.

The motivating case is the Stroop task, in which readers must respond to ink color while ignoring word meaning. Incongruent stimuli slow performance because the irrelevant word dimension competes with the goal-relevant color dimension. The task therefore provides a compact testbed for separating classical interference from an information-theoretic demonstration⁹.

This paper makes three contributions. First, it defines relevance as a goal-conditioned relation rather than a casual judgment by the experimenter. Second, it distinguishes statistical information from attention itself, avoiding the mistake of treating any predictive signal as useful

focus. Third, it applies the framework to a public Stroop dataset as a proof of concept, while explicitly limiting the claim to condition information preserved in response-time bins.

Methods

Research design

This study used a theoretical framework plus secondary quantitative analysis. The framework defines variables that could be used in future trial-level attention studies. The empirical portion reanalyzed an existing public Stroop dataset to show how one information-theoretic quantity can be estimated alongside traditional response-time statistics. Because the analysis used a public dataset rather than newly recruited participants, the result should be read as a proof of concept rather than a validated diagnostic measure.

Data source and sample

The proof-of-concept analysis used the public Stroop dataset made available by Daniel Lakens. The file contains 131 participant rows, with each row containing a participant identifier, congruent block completion time, incongruent block completion time, and year variable¹⁰.

No participant-level demographic variables were used. The dataset is block-level rather than trial-level, so it cannot isolate every relevant and irrelevant stimulus feature within each individual trial.

Variables and measurements

The classical behavioral variables were congruent block completion time, incongruent block completion time, and the within-participant interference difference. For the information-theoretic analysis, condition C had two states: congruent and incongruent. The response variable B_RT was a three-level response-time bin: Fast, Medium, or Slow.

Response-time bins were created from pooled response-time values across both conditions. Tertile cutoffs were used so that the bins contained approximately equal total counts. This binning choice was defined before computing mutual information, reducing the risk of choosing cutoffs that exaggerate the result.

Formal framework

The framework begins with a goal G , an action set A , a stimulus field S , and a loss function $L(a,s)$. A feature is treated as relevant only if changing that feature would change the optimal action, expected loss, or downstream task success under G . This definition must be made before analysis whenever possible; otherwise, relevance can become circular because any feature that predicts a response could be retroactively labeled relevant.

If trial-level data are available, relevant attentional throughput could be written as:

$$T_A(G) = I(S_{rel}; R | G) / \Delta t$$

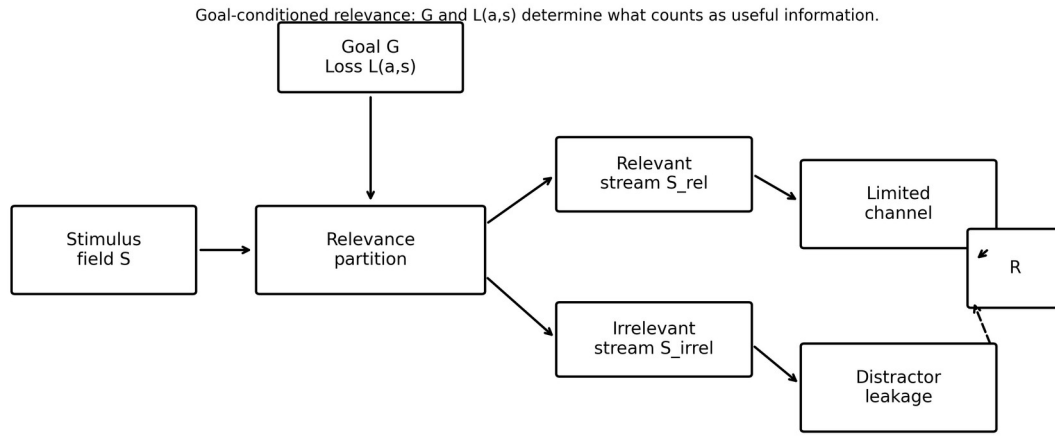
Here, $I(S_{rel}; R | G)$ is conditional mutual information between goal-relevant stimulus features and responses, conditioned on the goal, and Δt is a preregistered time window. Distractor leakage can be expressed as:

$$D(G) = I(S_{irrel}; R | G) / [I(S_{rel}; R | G) + \epsilon]$$

In this ratio, ϵ is a small positive constant in bits, such as 0.001 bits, used only to prevent division by zero. Any study using this ratio should report sensitivity to ϵ . A future effort-adjusted efficiency quantity could be written as:

$$E_A(G) = T_A(G) - \lambda C_{effort} - \gamma D(G)$$

In this expression, λ is the preregistered effort-cost weight and γ is the preregistered leakage-penalty weight. E_A is not estimated in this paper. The effort component would require additional measures such as pupil diameter, self-reported effort, heart-rate variability, or neural indicators of control cost^{13,14,15}.



The framework separates relevant throughput from leakage by irrelevant but response-predictive information.

Figure 1. Goal-conditioned attention channel. The goal G and loss function $L(a,s)$ enter the relevance-partition step as exogenous constraints rather than as products of the stimulus field. Relevant information enters the limited channel, while irrelevant information can still leak into response behavior.

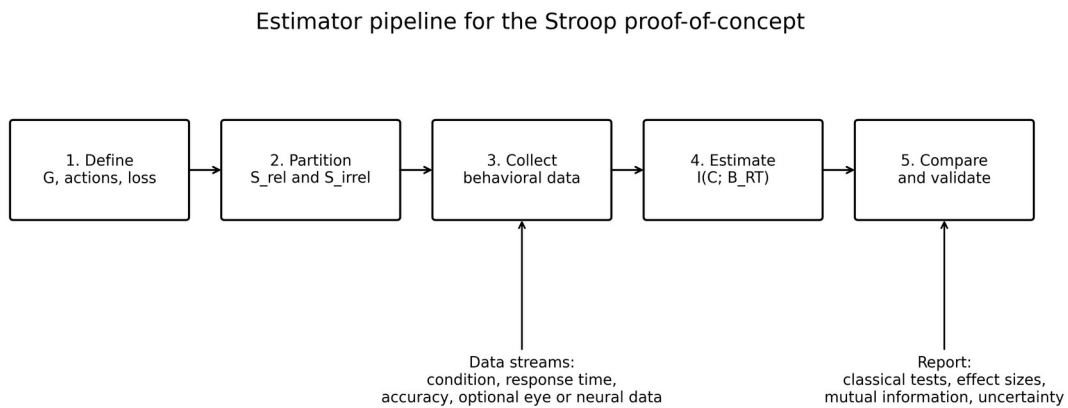


Figure 2. Estimator pipeline for the proof-of-concept. The pipeline separates goal definition, feature partitioning, data collection, information estimation, and validation so that the mathematics cannot silently replace experimental design.

Table 1. Operational definitions for the framework.

Element	Operational definition	Required pre-analysis decision
G	Task goal that defines success.	State the goal before analysis.
S_rel	Stimulus features that can change the optimal action or expected loss under G.	Define by task instructions, loss function, or preregistered expert coding.
S_irrel	Stimulus features not needed for the target goal, even if they are salient or behavior-predictive.	Do not relabel after seeing outcomes.
R	Measured response trace, such as response time, accuracy, gaze, or neural state.	Specify the response channel and sampling window.
D(G)	Distractor leakage ratio based on irrelevant information relative to relevant information.	Preregister ϵ and report sensitivity.
E_A(G)	Future efficiency score balancing throughput, effort, and leakage penalty.	Preregister λ and γ ; estimate effort with independent measures.

Data analysis

The traditional Stroop effect was analyzed with a paired-samples t-test comparing congruent and incongruent block completion times. Cohen's d_z was computed as t divided by the square root of n .

The information-theoretic calculation used the contingency table between condition C and response-time bin B_RT . Raw mutual information $I(C;B_RT)$ was computed in bits. Because finite samples can positively bias discrete information estimates, a Miller-Madow correction was also reported. A participant-level bootstrap with 100,000 resamples estimated uncertainty around $I(C;B_RT)^{11}$.

The demonstrated quantity, $I(C;B_RT)$, is not the same as full $T_A(G)$. Full T_A would require trial-level decomposition of relevant and irrelevant stimulus features. Continuous future datasets could use estimators such as k -nearest-neighbor mutual information, but those methods require more observations and stronger modeling assumptions than the present block-level dataset supports¹².

Ethical considerations

No new participants were recruited for this project. The analysis used a publicly available dataset with no personal identifying information in the analyzed file. Because this was a secondary analysis of deidentified public data, institutional review board approval was not required for the present manuscript. The broader discussion still treats attention data as ethically

sensitive because future attention measurement systems could involve behavioral logs, eye tracking, or neural recordings.

Results

Congruent blocks were completed faster than incongruent blocks. The mean congruent completion time was 15.10 s, and the mean incongruent completion time was 23.00 s. The mean interference effect was therefore 7.90 s. A paired-samples test showed a statistically reliable difference, $t(130) = 18.04$, $p < .001$, with Cohen's $d_z = 1.58$.

The response-time bin contingency table showed the expected ordering. Congruent observations were concentrated in the Fast bin, whereas incongruent observations were concentrated in the Slow bin. The raw mutual information between condition and response-time bin was $I(C;B_{RT}) = 0.404$ bits. The Miller-Madow corrected value was 0.399 bits. The bootstrap 95% interval for raw $I(C;B_{RT})$ was [0.324, 0.505] bits.

The contingency table also supported a simple majority-bin classifier. Assigning Fast to congruent, Medium to incongruent, and Slow to incongruent would correctly classify 208 of 262 observations, or 79.4%. This classifier is reported only as a descriptive comparison and not as an independent validation of the information-theoretic framework.

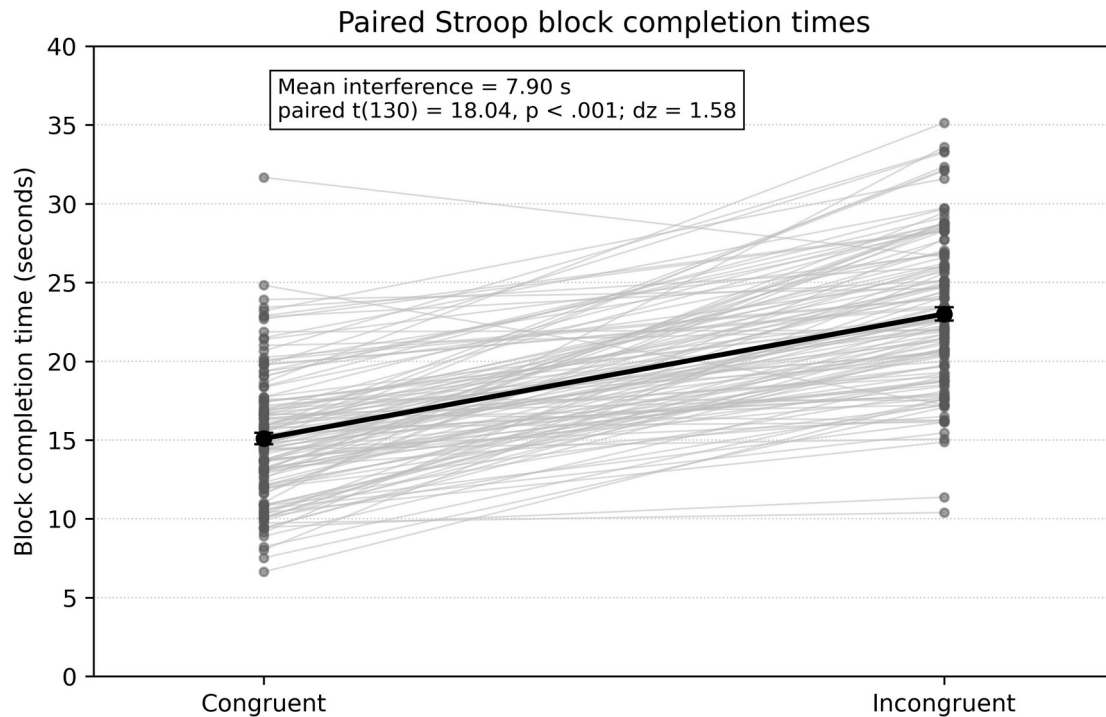


Figure 3. Paired Stroop block completion times. Each gray line represents one participant's congruent and incongruent block times; the black line shows the condition means. The mean interference effect was 7.90 s, paired $t(130) = 18.04$, $p < .001$, Cohen's $d_z = 1.58$.

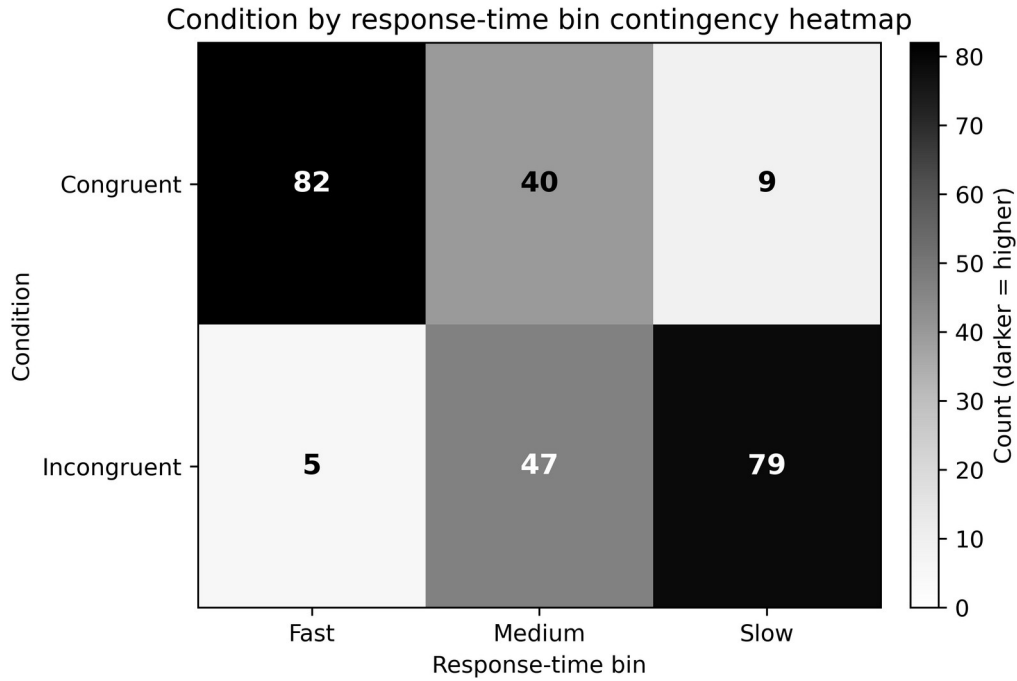


Figure 4. Condition by response-time bin contingency heatmap. Darker cells indicate higher counts. The dark upper-left and lower-right cells show that congruent observations cluster in the Fast bin and incongruent observations cluster in the Slow bin, consistent with the interference effect. Raw $I(C;B_{RT}) = 0.404$ bits; Miller-Madow corrected $I(C;B_{RT}) = 0.399$ bits.

Table 2. Stroop proof-of-concept statistics.

Quantity	Value	Interpretation
Mean congruent time	15.10 s	Faster baseline block completion
Mean incongruent time	23.00 s	Slower block completion under conflict
Mean interference	7.90 s	Within-participant Stroop slowing
Paired-samples test	$t(130) = 18.04, p < .001$; Cohen's $d_z = 1.58$	Classical evidence of interference
Raw $I(C;B_{RT})$	0.404 bits	Condition information preserved in response-time bin
Miller-Madow corrected $I(C;B_{RT})$	0.399 bits	Finite-sample bias-corrected information estimate
Bootstrap 95% interval	[0.324, 0.505] bits	Participant-level uncertainty for raw $I(C;B_{RT})$
Majority-bin classifier accuracy	79.4%	Descriptive agreement between condition and bin

Discussion

Key findings

The proof-of-concept supports the basic empirical expectation that Stroop interference appears in both a classical response-time analysis and an information-theoretic summary. The paired t-test shows that incongruent blocks are slower than congruent blocks. The contingency analysis shows that condition is strongly associated with response-time bin. These findings are internally consistent, but they do not prove that attention has a single bandwidth value.

The most important caution is terminological. Mutual information measures statistical dependence, not attention itself. A distraction can carry information if it predicts behavior, even when it damages the goal. For that reason, the framework treats information as useful only after relevance has been defined relative to G and $L(a,s)$.

Implications and significance

The framework is useful because it forces an analyst to specify what counts as goal-relevant information before claiming that a person is focused. In education, the target might be information that improves delayed problem solving. In medical image interpretation, the target might be lesion-relevant visual evidence. In an artificial-intelligence interface, the target might be information that improves task performance without increasing unnecessary cognitive offloading.

This distinction matters in environments designed to capture attention. A platform can maximize engagement while weakening task control. Conversely, a well-designed assistant may reduce irrelevant load while preserving the human role in goal selection, judgment, and decision-making. Cognitive offloading is therefore neither automatically harmful nor automatically beneficial; its value depends on whether it protects the limited channel or replaces the user's own evaluation of what matters¹⁶.

Limitations

The largest limitation is that the dataset is block-level. Because the data do not contain trial-by-trial stimuli, eye movements, accuracy, or neural activity, the analysis cannot compute $I(S_{rel}; R | G)$ directly. It computes $I(C; B_{RT})$, which is a simpler condition-response association. The manuscript therefore demonstrates an estimator style, not a complete measure of attentional bandwidth.

A second limitation is that relevance remains vulnerable to circularity if G is defined after seeing the data. To avoid this problem, future work should preregister the goal, action space, loss function, relevance partition, binning strategy, and validation outcomes before analysis. E_A also remains a future extension because effort cost, λ , and γ were not estimated in the present dataset.

Recommendations for future research

The strongest next version would narrow to one application, such as AI-assisted learning, and run a preregistered trial-level study. Such a study should collect task stimuli, responses, accuracy, response times, subjective effort, and at least one independent attention measure such as eye tracking or EEG. It should then test whether the proposed information metrics predict delayed performance beyond accuracy, mean response time, and time-on-task.

Future studies should also compare the framework against simpler baselines. If $I(S_{rel};R | G)$, $D(G)$, or E_A fails to predict outcomes beyond ordinary measures, the framework should be rejected or revised. Mathematical elegance is not evidence by itself.

Closing thought

The goal is not to optimize engagement metrics. It is to protect the limited channel through which people decide what matters.

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